Reg. No. | | | | | | | | | | | [17CS4DS01]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

SCIENTIFIC PROGRAMMING USING R

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions. Choosing at least **TWO** questions from each part.
2. Any missing data may be suitably assumed.

PART – A

1.	a.	i.	When did the R programming language come into existence and why was it called as "R"?	03
		ii.	Give three commands to assign *4* to the variable *var*.	03
		iii.	Write code for generating a sequence as *(1 2 3 4 2 3 4 5 3 4 5 6 4 5 6 7)* with *rep()* and *seq()* functions.	03
	b.	i.	Generate a variable having 5 names as string along with three *NA* values and also check if the variable has *missing* values.	2.5
		ii.	Write code for a matrix having 12 random values in the range 1 to 20 and fill the elements row-wise. Further to find all the elements from its 3rd column.	2.5
2.	a.	i.	What are the codes for installing and calling a package in R script?	03
		ii.	Generate a vector containing all integers from 1 to 100 that are not divisible by 2, 3, or 5.	03
		iii.	Write code to produce a diagonal matrix with elements 1:4 and get its inverse also.	03
	b.	i.	Produce a vector having a set of four elements *(1, 2, 3, 4)* five times collectively and five times each of the elements.	2.5
		ii.	Write code to produce a matrix of *3X4* with all its elements as missing values. Further to find its object type as either matrix or vector.	2.5
3.	a.		Write code to list down all the objects and further to delete a particular variable from the environment and also to clear the work space.	08
	b.		Write an R program to display a factorial for 1:10 in tabular form using loop with *cat(), paste()*, and *format()* commands.	06
4.	a.	i.	Solve $\lim_{x\to 2}\frac{x^2-4}{x-2}$ by the packages "rSymPy" and "Ryacas" and write about the outcomes from both the packages.	04
		ii.	Create 5 variables for 10 employees as emp_id, name, age, sex (female, male), and salary and put them in a data frame. Further obtain the	04

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summary for the data frame. Further factorize the sex variable as (0, 1) and again write code for summary of the data frame. **06**

iii. Name the three most important OOP systems from R and explain about them briefly.

PART - B

5. a. i. Write a small introduction about object oriented programming S3. **05**

ii. Tabulate function f (f=x^3-2*x^2+3) values and its derivative values in the range (-3, 3) with the increment value of 0.1 using R statements. **05**

b. Perform triple integration expression f = x for the limits (0, 0, 0) and (1, 1, 1) using adaptIntegrate() function from 'cubature' package. **04**

6. a. i. Implement the functions used for merging of data frames horizontally and vertically in R. Give an example for each. **05**

ii. What are the differences between lapply() and sapply() functions? Differentiate with examples. **05**

b. Name the data structures used in R. Express examples for each. **04**

7. a. i. Explain about arrays in R. Create an array with two vectors vec1 = 45, 32, 35, 26, -34; and vec2 = -23, 28, 29, 80, 12, 15 with dimension 3 x 3 x 2. **05**

ii. Write a function to generate the following sequence: 1, 9, 36, 100, . . . up to 10 terms. **05**

b. Implement mean, median and mode functions with examples of R codes. **04**

8. a. i. Explain about lists in R. Create a list with variables colors = red, blue, green, yellow; digits = even numbers in the range 1 to 20; and binomialValues = TRUE, FALSE. **05**

ii. Write a function to generate the following sequence: 1, 5, 14, 30 . . . , up to 10 terms. **05**

b. Describe scan() and readline() functions for getting the output from the user and system storage as well with examples. **04**